Midterm. Row 2. Solutions

Mathematical Analysis
1st Year

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Bachelor in Computer Science

Midterm Test

1. Let $A \subseteq \mathbb{R}$. Prove that $\text{int}A \subseteq A \subseteq \text{cl}A$. Give an example with strict inclusions. 1p

- $\text{int} A = \{x \in \overline{\mathbb{R}} \mid \exists V \in \mathcal{V}(x),\ V \subseteq A\}$, $\text{cl} A = \{x \in \overline{\mathbb{R}} \mid \forall V \in \mathcal{V}(x),\ V \cap A \neq \emptyset\}$. 0.25p
- Let $x \in \text{int} A \Rightarrow \exists V \in \mathcal{V}(x),\ V \subseteq A$. Since $x \in V$, we have that $x \in A$. 0.25p
- Let $x \in A$. $\forall V \in \mathcal{V}(x)$, $x \in V$, so $V \cap A \neq \emptyset \Rightarrow x \in \text{cl}A$. 0.25p
- For $A = [0,1)$, $\text{int} A = (0,1)$, $\text{cl}(A) = [0,1]$. Any example of A not open, nor closed is ok. 0.25p

2. Find the value of $\lim\limits_{n\to\infty} \dfrac{1}{2n+1} + \dfrac{1}{2n+2} + \ldots + \dfrac{1}{3n}$. 1p

- $$\sum_{k=1}^{n} \frac{1}{2n+k} = \frac{1}{n}\sum_{k=1}^{n} \frac{1}{2+\frac{k}{n}} \xrightarrow{0.5p} \int_0^1 \frac{1}{2+x}\,dx = \ln(x+2)\Big|_0^1 = \ln\frac{3}{2}. \quad 0.5p$$
- Or we can use the sequence $c_n = 1 + \frac{1}{2} + \ldots + \frac{1}{n} - \ln n \to \gamma$
$$\sum_{k=1}^{n} \frac{1}{2n+k} = c_{3n} - c_{2n} + \ln 3n - \ln 2n = \underbrace{c_{3n} - c_{2n}}_{\to 0} + \ln\frac{3}{2} \longrightarrow \ln\frac{3}{2}.$$

3. Study the convergence of the following: 3p

(a) $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$. (b) $\displaystyle\int_0^{\pi} \frac{1}{\sqrt{\sin x}}$. (c) $\displaystyle\sum_{n=1}^{\infty} \frac{1\cdot 3 \cdot \ldots \cdot (2n-1)}{2^n n!}$.

(a) • Using the integral test, the series has the same nature as 0.5p
$$\int_2^{\infty} \frac{1}{x(\ln x)^2}\,dx \underset{u=\ln x,\ du=\frac{1}{x}dx}{=} \int_{\ln 2}^{\infty} \frac{1}{u^2}\,du = -\frac{1}{u}\Big|_{\ln 2}^{\infty} = 0 + \frac{1}{\ln 2},$$ hence it is convergent. 0.5p

• Or using the Cauchy condensation test: $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ has the same nature as $\sum 2^n \frac{1}{2^n(\ln 2^n)^2} = \sum \frac{1}{n^2 \ln^2 2}$ convergent

(b) Improper integral at 0 and π where $\sqrt{\sin x} = 0$.
$$\int_0^{\pi} \frac{1}{\sqrt{\sin x}}\,dx = \int_{0+0}^{\frac{\pi}{2}} \frac{1}{\sqrt{\sin x}}\,dx + \int_{\frac{\pi}{2}}^{\pi-0} \frac{1}{\sqrt{\sin x}}\,dx.$$

Recall that $\displaystyle\int_{a+0}^{b} \frac{1}{(x-a)^p}\,dx$, $\displaystyle\int_{a}^{b-0} \frac{1}{(b-x)^p}\,dx$ converges if $p<1$, diverges if $p \geq 1$. 0.5p

Since $\lim\limits_{x\to 0} \frac{\sin x}{x} = 1$, $\lim\limits_{x\to 0} \dfrac{\frac{1}{\sqrt{\sin x}}}{\frac{1}{\sqrt{x}}} = 1$, hence $\displaystyle\int_{0+0}^{\frac{\pi}{2}} \frac{1}{\sqrt{\sin x}}\,dx$ "like" $\displaystyle\int_{0+0} \frac{1}{\sqrt{x}}\,dx$, convergent. 0.25p

Also, $\lim\limits_{x\to\pi} \frac{\sin x}{\pi - x} = \lim\limits_{x\to\pi} \frac{\sin(\pi - x)}{\pi - x} = 1$, hence $\displaystyle\int_{\pi/2}^{\pi-0} \frac{1}{\sqrt{\sin x}}\,dx$ "like" $\displaystyle\int_{\pi/2}^{\pi-0} \frac{1}{\sqrt{x}}\,dx$, convergent. 0.25p

Conclusion: $\displaystyle\int_0^{\pi} \frac{1}{\sqrt{\sin x}}\,dx$ is convergent.

(c) $x_n = \frac{1\cdot 3\cdot\ldots\cdot(2n-1)}{2^n n!}$, $\frac{x_{n+1}}{x_n} = \frac{\cancel{1\cdot 3\cdots(2n-1)}(2n+1)}{2^{n+1}(n+1)\cancel{!}}\cdot\frac{\cancel{2^n n!}}{\cancel{1\cdot 3\cdots(2n-1)}} = \frac{2n+1}{2(n+1)} = \frac{2n+1}{2n+2} \to 1$. 0.5p

0.25p Raabe-Duhamel test: $n\left(\frac{x_n}{x_{n+1}}-1\right) = n\left(\frac{2n+2}{2n+1}-1\right) = n\cdot\frac{1}{2n+1} \to \frac{1}{2} < 1 \to$ divergent. 0.25p

4. Find the Taylor series around 0 and its radius of convergence for each of the following:

(a) $\ln(1+x)$. 1p (b) $1/(1+x^2)$. 1p

(a) $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n \longrightarrow$ Taylor series around 0. 0.25p

$f(x) = \ln(1+x)$, $f(0)=0$; $f'(x) = \frac{1}{x+1}$, $f'(0)=1$; $f''(x) = -\frac{1}{(x+1)^2}$, $f''(0)=-1$.

$f^{(3)}(x) = \frac{2}{(x+1)^3}$, $f^{(3)}(0) = 2$, $f^{(4)}(x) = \frac{-2\cdot 3}{(x+1)^4}$, $f^{(4)}(0) = -3!$

$f^{(n)}(x) = (-1)^{n+1}\frac{(n-1)!}{(x+1)^n}$, $f^{(n)}(0) = (-1)^{n+1}(n-1)!$, $n\geq 1$. 0.5p

Hence $\ln(1+x) = f(x) = f(0) + \sum_{n=1}^{\infty}\frac{f^{(n)}(0)}{n!}x^n = \sum_{n=1}^{\infty}\frac{(-1)^{n+1}(n-1)!}{n!}x^n = \sum_{n=1}^{\infty}(-1)^{n+1}\frac{x^n}{n}$.

- Radius of convergence: $R = \frac{1}{L}$, $L = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{n}{n+1}\right| = 1 \to R = 1$. 0.25p
- Or we can consider $\frac{1}{1+x} = \sum_{n=0}^{\infty}(-x)^n$ and integrate: $\ln(1+x) = \sum_{n=0}^{\infty}(-1)^n\frac{x^{n+1}}{n+1} = \sum_{n=1}^{\infty}(-1)^{n-1}\frac{x^n}{n}$
$\forall x\in(-1,1)$

(b) The simplest solution here is to use the geometric series $\sum_{n=0}^{\infty}x^n = \frac{1}{1-x}$, $\forall x\in(-1,1)$.

$\frac{1}{1+x^2} = \sum_{n=0}^{\infty}(-x^2)^n = \sum_{n=0}^{\infty}(-1)^n x^{2n}$, $\forall x\in(-1,1)$. 0.75p

Radius of convergence: $R = \frac{1}{L}$, where $L = \lim_{n\to\infty}\left|\frac{a_{2n+2}}{a_{2n}}\right| = 1 \to R = 1$. 0.25p

Remark: for writing the Taylor series formula and computing $f'(0)$, $f''(0)$ 0.5p

5. Consider the power series $\sum_{n=1}^{\infty}\frac{x^n}{n^2+n}$.

(a) Find its radius of convergence and its convergence set. 1p

(b) Find its sum for any x in the convergence set. 1p

(a) $\sum_{n=1}^{\infty}\underbrace{\frac{1}{n(n+1)}}_{a_n}x^n$. Radius of convergence $R = \frac{1}{L}$, where $L = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = 1 \to R = 1$. 0.5p

- for $x=1$: $\sum_{n=1}^{\infty}\frac{1}{n^2+n}$ has the same nature as $\sum_{n=1}^{\infty}\frac{1}{n^2}$, which is convergent. 0.25p
- for $x=-1$: $\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2+n}$ is convergent by the Leibniz test since $\frac{1}{n^2+n}\to 0$. 0.25p

(b) $\sum_{n=1}^{\infty}\frac{x^n}{n^2+n} = \sum_{n=1}^{\infty}\left(\frac{1}{n}-\frac{1}{n+1}\right)x^n = \sum_{n=1}^{\infty}\frac{x^n}{n} - \sum_{n=1}^{\infty}\frac{x^n}{n+1} = \sum\frac{x^n}{n} - \frac{1}{x}\sum_{n=1}^{\infty}\frac{x^{n+1}}{n+1}$

$\sum\limits_{n=1} \quad n^2+n$

(b) $\displaystyle\sum_{n=1}^{\infty} \frac{x^n}{n(n+1)} = \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1}\right) x^n = \sum_{n=1}^{\infty} \frac{x^n}{n} - \sum_{n=1}^{\infty} \frac{x^n}{n+1} = \sum_{n=1}^{\infty} \frac{x^n}{n} - \frac{1}{x} \sum_{n=1}^{\infty} \frac{x^{n+1}}{n+1}$. 0.25p

- $\displaystyle\left(\sum_{n=1}^{\infty} \frac{x^n}{n}\right)' = \sum_{n=1}^{\infty} \left(\frac{x^n}{n}\right)' = \sum_{n=1}^{\infty} x^{n-1} = \sum_{n=0}^{\infty} x^n = 1+x+x^2+\dots = \frac{1}{1-x}, \quad \forall x \in (-1,1).$

Integrating we obtain that $\displaystyle\sum_{n=1}^{\infty} \frac{x^n}{n} = -\ln(1-x), \quad \forall x \in (-1,1)$. 0.25p

- $\displaystyle\sum_{n=1}^{\infty} \frac{x^{n+1}}{n+1} = \sum_{n=2}^{\infty} \frac{x^n}{n} = -\ln(1-x) - x, \quad \forall x \in (-1,1).$

- We get that $\displaystyle\sum_{n=1}^{\infty} \frac{x^n}{n(n+1)} = -\ln(1-x) + \frac{\ln(1-x)}{x} + 1 =: g(x), \quad \forall x \in (-1,1)$. 0.25p

- Notice that when $x=0$ we get $0 = g(0) = \lim\limits_{x\to 0} g(x) \overset{l'H}{=} -0-1+1=0$.

- Notice that g can also be defined for $x = \pm 1$, where the series converges, hence we have that $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = g(1) = \lim_{x \nearrow 1} \Big[\underbrace{\ln(1-x)\cdot\frac{1-x}{x}}_{\to 0} + 1\Big] = 1.$

0.25p

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n(n+1)} = g(-1) = -\ln 2 - \ln 2 + 1 = 1 - 2\ln 2.$$